11.4 Modeling with Area

Geometry 11.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- find and compare population densities.
- find areas of real-life regions and convert between square units.
- find the rectangle with the greatest area for a given perimeter.

Vocabulary · fill in the missing word

Population density — a _____ of how many people live within a given area

Square unit — the _____ of a square with sides of one unit

Part 1 · Discovery: how crowded is our classroom?

Measure the classroom. Count the people in it.

Length (ft)	Width (ft)	Area (ft^2)	People	People per ft^2	ft^2 per person

At a concert, each person needs about 5 ft^2. How many people could stand in our room? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Population density = _____ ÷ _____

Example 1

About 2,240 people live in a town that covers 4.5 square miles. Find the population density. Round to the nearest tenth.

Area first: when a region is a circle or polygon, find its _____ with a formula, then divide.

Example 2

About 52,000 people live within 6 miles of a city's center. Find the population density of that circular region. Round to the nearest tenth.

Example 3

County A has 245,000 people in 610 square miles. County B has 82,000 people in 175 square miles. Which county is more densely populated?

Square units: 1 yd^2 = _____ ft^2 (3 ft × 3 ft), so divide square feet by 9 to get square yards.

Example 4

A lawn has an area of 1,800 square feet. Sod costs $3.50 per square yard. How much will the sod cost?

Greatest area: for a fixed perimeter, the rectangle with the greatest area is a _____.

Example 5

You have 64 feet of fencing for a rectangular garden. What dimensions give the greatest area? What is that area?

Part 3 · Partner practice

Partner 1

A room is 15 feet by 12 feet. Carpet costs $27 per square yard. How much will it cost to carpet the room?

Partner 2

A circular park with a radius of 0.4 mile has about 1,500 trees. Find the number of trees per square mile. Round to the nearest whole number.